

B.TECH. COMPUTER SCIENCE AND ENGINEERING**DESIGN AND ANALYSIS OF ALGORITHMS****Course Code** : 23CSE402**Credits** : 3**L:T:P:E** : 2:0:2:2**Contact Hours** : 60**Prerequisite** : Data Structures**Course Objectives**

The objective of the course is to

1. Analyse the asymptotic performance of algorithms.
2. Demonstrate a familiarity with major algorithms and data structures.
3. Apply important algorithmic design paradigms and methods of analysis.
4. Synthesize efficient algorithms in common engineering design situations.

Course Outcomes

At the end of the course, students will be able to

Course Outcomes	Description	Bloom's Taxonomy Level
CO1	Discuss the Characteristics of algorithm, the notation of algorithms.	Understanding (2)
CO2	Illustrate the sorting, searching, graph traversal problem and divide and conquer design techniques.	Understanding (2)
CO3	Solve the given real time problems through an greedy appropriate design strategy.	Applying (3)
CO4	Solve the combinatorial optimization problems which efficiently estimate the search space's lower and upper limits of branches.	Applying (3)
CO5	Illustrate the other design paradigms and apply.	Applying (3)
CO6	Analyze the NP Complete problems.	Analyzing (3)

B.TECH. COMPUTER SCIENCE AND ENGINEERING**Course Contents****Module -1 Introduction****[09 hours]**

Algorithm Definition, Characteristics of an algorithm, Algorithm as a technology, Performance Analysis: Space complexity, Time complexity. Asymptotic Notations: Big-Oh notation, Omega notation, Theta notation.

Module -2**[09 hours]**

Searching and Sorting : Linear Search, Insertion sort

Divide and Conquer: Binary Search, Merge sort, Quick sort.

Graph Traversal: Breadth First Search and Traversal, Depth First Search and Traversal.

Module -3**[09 hours]**

Greedy method:

The General Method, Minimum cost spanning trees, Single source shortest paths.

Dynamic Programming:

The General method, all pairs shortest path, traveling salesperson problem, Reliability Design

Backtracking:

The General method, the n-Queens problem, Subset-sum Problem.

Module -4**[09 hours]**

Branch and Bound:

The method: FIFO Branch and Bound, LC Branch and Bound, 0/1 knapsack problem, Traveling salesperson, Efficiency considerations.

Module 5:**[09 hours]**

NP-Complete and NP-Hard Problems:

P class, NP class, NP hard class, NP complete class – their interrelationship, Satisfiability problem, Cook's theorem (Statement only), Clique decision problem

TEXT BOOKS:

1. Fundamentals of computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekaran, 2nd Edition, 2014, Universities Press
2. Introduction to the Design and Analysis of Algorithms, Anany Levitin: 2nd Edition, 2009. Pearson

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REFERENCES:

1. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI
2. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education).

Lab Course Contents:

Total Hrs : 15 hours

Lab Exp #	Details
1	a. Demonstrate Iterative approaches and display the running time of the program. b. Demonstrate Recursion and display the running time of the program.
2	a. Demonstrate Linear Search in Single and Multi-Dimensional Array. b. Implement Divide and conquer technique: i. Search an element using Binary Search in sorted and unsorted list. ii. Sort a given sent of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted. The elements can be read from a file or can be generated using the random number generator. iii. Sort a given sent of elements using the Merge sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted. The elements can be read from a file or can be generated using the random number generator. c. Print all the nodes reachable from a given starting node in a digraph using BFS and DFS method.
3	a. Demonstrate Minimum Cost Spanning Trees. b. Implement Single Source Shortest Paths algorithm. c. Implement All-Pairs Shortest Path Problem using Floyd's algorithm. d. Demonstrate Backtracking technique i. n-Queens problem (n = 4, n = 8) ii. Solving Sudoku puzzle.
4	a. Demonstrate 0/1 knapsack problem using Least Cost Branch and Bound. b. Demonstrate Travelling salesperson problem using Branch and Bound.
5	a. Find all cliques of size K in an undirected graph. b. Given a small graph with N nodes and E edges, the task is to find the maximum clique in the given graph. i.e., when an edge is added to the present list, check that by adding that edge to the present list, does it still form a clique or not, if the vertices are added until the list does not form a clique. Then, the list is backtracked to find a larger subset which forms a clique using Recursion (Maximal Clique Problem).

